MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID (MICROWAVE)

TYPE 7588

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, metal-ceramic
See figure 1
Mounting position: Any
Weight: 0.3 ounce (8.5 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	Eb	Ec	Egk	Ehk	Rk
Unit:	V	Vdc	Vdc	Vdc	Vdc	Ohms
Maximum:	6.6	300	---	0	+50	---
Minimum:	6.0	---	-50	---	-50	---
TEST CONDITIONS:	6.3	200	+6	---	---	270

ABSOLUTE RATINGS:

Parameter:	Rg	Ik	Pp	TE	Barometric pressure, reduced
Unit:	Meg	mAdc	W	°C	mmHg
Maximum:	0.025 fixed bias 0.1 cathode bias	30	5.5	250	8
Minimum:	---	---	---	---	---
TEST CONDITIONS:	---	---	---	---	---

GENERAL:

Qualification - Required

Ⓔ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
(E)	Qualification inspection								
1002	Barometric pressure, reduced	4	Voltage = 300 Vac; pressure = 8 ±1 mmHz	---	---	---	---	---	---
1031	Sweep-frequency vibration	9	Ebb = 250 Vdc; Rp = 2,000 ohms; Ec = 0, Rk = 68 ohms, 10 G; F = 100 to 2,000 Hz	---	---	Ep	---	75	mVac
1031	Sweep-frequency vibration fatigue	10	Eb = Ec = 0; Ef = 6.3 V	---	---	---	---	---	---
---	Sweep-frequency vibration fatigue test end points:								
1031	Low-frequency vibration	-		---	---	Ep	---	50	mVac
1336	Heater-cathode leakage	-		---	---	Ihk	---	30	µAdc
1306	Transconductance (1)	-		---	---	Sm	29,000	---	µmhos
1301	Heater current	-		---	---	If	370	440	mA
	Quality conformance inspection, part 1								
1301	Heater current	-		0.65	II	If	370	430	mA
1336	Heater-cathode leakage	-		0.65	II	Ihk	---	20	µAdc
1266	Total grid current	-	Ecc = -1.0 Vdc; Rg = 0.1 Meg; Rk = 22 ohms	0.65	II	Ic	---	-0.5	µAdc
1256	Electrode current (anode)	-	Ec = 0; Rk = 22 ohms	0.65	II	Ib	17	33	mAdc
1306	Transconductance (1)	1		0.65	II	Sm	35,000	55,000	µmhos
1261	Electrode voltage (grid)	-	Rk = 0; Ec/Ib = 100 µAdc	0.65	II	Ec	---	-8.0	Vdc
---	Noise figure	2	Ebb = 265 Vdc; Ec = 0; Rb = 3,300 ohms; Cb = 10 µF; Rk = 22 ohms	0.65	II	NF	---	4.8	dB
1201	Short and discontinuity detection	-		0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes	-		---	---	R	50	---	Meg
1306	Transconductance (2)	1	Ef = 5.7 V	---	---	ΔSm Ef	---	20	%
1316	Amplification factor	1		---	---	Mu	140	210	---
1266	Grid emission	3	Ef = 7.0 V; Ecc = -15 Vdc, Rk = 0, Rg = 0.1 Meg	---	---	Ic	0	-2.0	µAdc
2210	Grid recovery	6	Ebb = 250 Vdc, Ec/Ib = 10 mAdc, Rp = 0.01 Meg	---	---	ΔIb Δib	--- ---	1.0 2.0	mAdc ma
1331	Direct-interelectrode capacitance	-		---	---	Cgp Cin Cout Chk	2.1 5.1 0.05 1.9	3.5 8.3 0.1 3.3	pF pf pF pF

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued								
---	Envelope strain	5		---	---	---	---	---	---
Ⓛ 1031	Low-frequency vibration	8	Ebb = 250 Vdc, Rp = 2,000 ohms; Ec = 0, Rk = 68 ohms; 15 G; F = 40 Hz	---	---	Ep	---	25	mVac
	Quality conformance inspection, part 3								
1506	Heater-cycling life	7	Ef = 7.0 V; Ehk = +70 Vdc; Rk = 0; Ec = Eb = 0	---	---	---	---	---	---
---	Heater-cycling life-test end point:								
1336	Heater-cathode leakage	-		---	---	Ihk	---	40	µAdc
1516	Stability life	-	Eb = 200 Vdc; Ecc = +6 Vdc; Ehk = -70 Vdc; Rk = 270 ohms; Rg = 0.1 Meg	2.5	Code F	---	---	---	---
---	Stability life-test end point: (2 and 20 hours)								
---	Change in transconductance of individual tubes	-		---	---	ΔSm_t	---	15	%
1501	Intermittent life	-	Group A, stability life-test conditions, t = 1,000 hours	---	---	---	---	---	---
---	Intermittent life-test end points:								
1301	Heater current	-		---	---	If	370	440	mA
1266	Total grid current	-		---	---	Ic	---	-0.5	µAdc
1306	Transconductance (1)	-		---	---	Sm	29,000	---	µmhos
---	Change in transconductance (1) of individual tubes			---	---	ΔSm_t	---	30	%
1306	Transconductance (2)	-		---	---	ΔSm_{Ef}	---	25	%
---	Noise figure	-		--	---	NF	---	6.0	dB
1336	Heater-cathode leakage	-		---	---	Ihk	---	20	µAdc
1211	Insulation of electrodes	-		---	---	R	20	---	Meg
Ⓔ 1041	Shock	11	Hammer angle = 30°, Ehk = 100 Vdc; Eb = 250 Vdc, Rk = 68 ohms; Ec = 0	---	---	---	---	---	---
	Shock test end points:								
1031	Low-frequency vibration	-		---	---	Ep	---	50	mVac
1336	Heater-cathode leakage	-		---	---	Ihk	---	30	µAdc
1306	Transconductance (1)	-		---	---	Sm	29,000	---	µmhos
1301	Heater current	-		---	---	If	370	440	mA

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NOTES:

1. In performing this test, the cathode-by pass capacitor shall be charged to +6 volts prior to insertion of the tube into the test socket to prevent high initial surge current.

2. Noise figure test.

   (a) Definition. Noise figure shall be defined as:

   $F = 10 \log \frac{Si/Ni}{So/No}$ where

   F is noise figure expressed in dB.
   Si/Ni is signal to noise ratio at input of tube under test.
   So/No is signal to noise ratio at output of tube under test.

   (b) Test procedure. Tube under test shall be operated in a grounded-grid amplifier circuit at 200 ±10 MHz. The input circuit shall be adjusted for a noise match and tuned for a minimum noise figure. The output circuit shall be tuned and adjusted for a power match. External feedback around the tube under test, except for that required for neutralization or that which is unavoidable in a well designed circuit, shall not be used.

   A signal source which has known output constants over the bandwidth of the tube under test, such as a noise diode, gas discharge tube, or known resistance at a known temperature, shall be used in measuring the noise figure of the tube under test.

   The bandwidth of the tube under test circuit shall be at least twice that of the overall system used in measuring noise figure.

3. Prior to this test, tubes shall be preheated a minimum of 5 minutes at the conditions indicated below. Test within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef	Ecc	Eb	Rk	Rg
V	Vdc	Vdc	Ohms	Meg
7.0	0	250	50	0.1

4. The tube shall be tested as specified in method 1002, with the exception that the specified voltage shall be applied between the anode and grid.

5. Tubes shall be tested as specified in method 2126, except that they shall first be immersed in water at not more than 5°C for 15 seconds and immediately thereafter subjected to the standard temperature cycle specified in method 2126.

6. Grid recovery. Test shall be performed as follows:

   (a) With TUT operating under specified conditions, adjust Ec for Ib = 10 mAdc.
   (b) Apply 3 volt pulse, 20 µs duration, 60 pps to the grid.
   (c) With application of pulse, measure undershoot (Δib) and change in average current (ΔIb).

7. The heater-cycling life-test sample shall consist of 15 tubes and no tube failure shall be permitted. In the event of rejection of the first sample due to failure of 1 tube, a second sample of 15 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed one.

8. Low-frequency vibration. Test shall be performed as follows: The tube shall be vibrated with simple harmonic motion in each of two planes; first, parallel to the cylindrical axis; second, perpendicular to the cylindrical axis and perpendicular to major axis of terminal lug.

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NOTES: -Continued

9. Sweep-frequency vibration. Test shall be performed as follows:

   (a) The frequency shall be increased from 100 to 2,000 Hz with approximately logarithmic progression in 3 ±1 minutes. The return sweep (2,000 to 100 Hz) is not required.
   (b) The tube shall be vibrated with simple harmonic motion in each of two planes; first, parallel to the cylindrical axis; second, perpendicular to the cylindrical axis and parallel to a line through the major axis of a terminal lug. At all frequencies from 100 to 2,000 Hz the total harmonic distortion of the acceleration waveform shall be less than 5 percent.
   (c) The peak acceleration shall be maintained at 10 ±1.0 G throughout the test.
   (d) The value of the alternating voltage, Ep, produced across the resistor, Rp, as a result of the vibration shall be measured with a suitable device having a response to the RMS value of the voltage to within ±0.5 dB of the response at 400 Hz or the frequency range of 100 to 3,000 Hz, and having a bandpass filter with an attenuation rate of 24 dB per octave below the low-frequency cut-off point of 50 Hz and above the high-frequency cut-off point of 5,000 Hz. The meter shall have a dynamic response characteristic equivalent to or faster than a VU meter (operated in accordance with Standard ASA No. C16.5-1954).

10. Sweep-frequency vibration fatigue. Test shall be performed as specified in method 1031, except that the tubes shall be vibrated for a total of 6 hours, that is, 3 hours in each of two directions; first, parallel to the cylindrical axis; second, perpendicular to the cylindrical axis and parallel to a line through the major axis of a terminal lug.

(E) 11. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

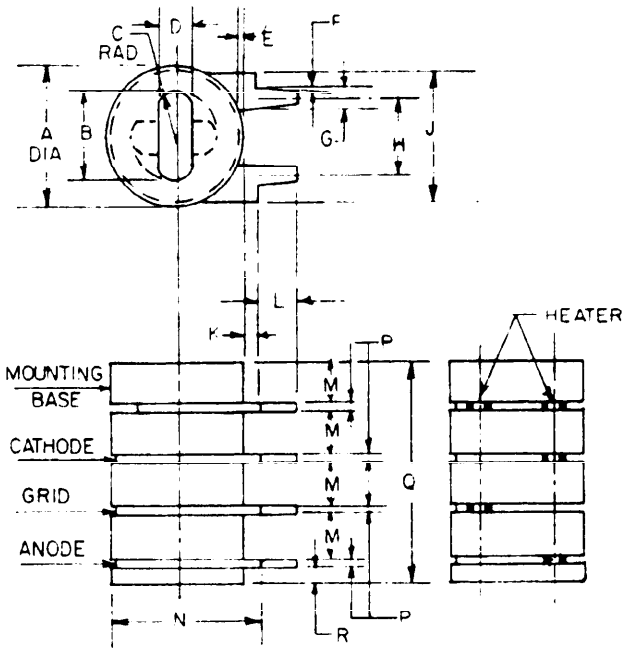
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 11

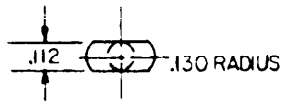
Preparing activity:
Navy - EC

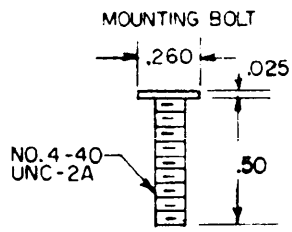
Agent:
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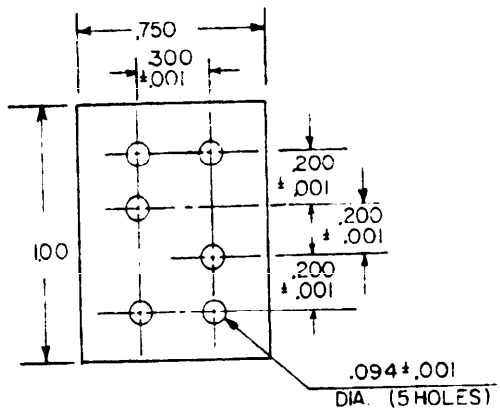
Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A	.535(13.59)	.565(14.35)
B	.290 (7.37)	.310 (7.87)
D	.115 (2.92)	.145 (3.68)
G	.080 (2.03)	.086 (2.18)
N	.595(15.11)	.635(16.13)
P	.025 (.64)	.031 (.79)
Q	.860(21.84)	.890(22.61)
Quality conformance inspection, part 2		
C	.140 (3.56)	.160 (4.06)
H	.290 (7.37)	.310 (7.87)
J	.490(12.45)	.510(12.95)
L	.145 (3.68)	.155 (3.94)
M	.170 (4.32)	.180 (4.57)
R	.070 (1.78)	.080 (2.03)
Reference dimensions		
E	.025 (.64)	
F	.006 (.15)	
K	.065 (1.65)	

INCHES	MM
.025	.64
.112	2.84
.130	3.30
.260	6.60
.50	12.70

NOTE:
All terminals shall be so alined that they are capable of being inserted into a gage .125(3.18 mm) thick having 5 holes .094(2.39 mm) ± .001(.03 mm) in diameter as shown in figure 2.

FIGURE 1. Outline drawing of electron tube type 7588.

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INCHES	MM
.001	.03
.094	2.39
.200	5.08
.300	7.62
.750	19.05
1.00	25.40

FIGURE 2. Terminal-position gage.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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